Bracketing eigenvalues from a tension curve

The mathematics of `chladni`'s v1 bracket search

chladni development notes

Abstract

The method of particular solutions turns the Dirichlet eigenvalue problem on a planar domain into the problem of finding the local minima of a scalar *tension* curve $\sigma_1(\lambda)$, the smallest generalized singular value of a matrix pencil. This note develops the algorithm `chladni` uses for the first half of that search: given a window $[a, b]$ and a requested relative precision prec, produce a set of intervals each believed to contain one eigenvalue, and then drive each down to machine accuracy. Four ideas carry the design. (i) Near an eigenvalue σ_1 is a cone with a rounded bottom, whose two parameters — a floor ε belonging to the *basis* and a wall slope c belonging to the *domain* — are independent, and whose ratio ε/c is the resolution limit. (ii) The wall slope scales as $c(\lambda) = C/\sqrt{\lambda}$, which makes $k = \sqrt{\lambda}$ the *unique* power of λ in which one Lipschitz bound serves a whole window; this change of variable is worth more than every algorithmic change in the design combined. (iii) A Lipschitz bound converts samples into intervals *certified to contain no eigenvalue*, and the cost of certifying a cone's wall is a geometric ladder whose ratio depends only on the safety factor — which fixes the cost of the method and retires a family of attempts to make it cheaper. (iv) One evaluation returns the whole singular-value vector, so σ_2 is free, and σ_2/c is a distance meter that is accurate exactly where the decision is hard. We derive each, state honestly what is proved and what is only measured, and record what changed when the algorithm was ported from the repository it was designed in.

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1 What the bracketing stage is asked for

1.1 From the pencil to a scalar curve

Let $\Omega \subset \mathbb{R}^2$ be a bounded planar domain and consider the Dirichlet eigenvalue problem $-\Delta u = \lambda u$ in Ω, $u = 0$ on $\partial\Omega$. The method of particular solutions (MPS) builds, at each trial value λ, a basis $\psi_1(\cdot;\lambda), \ldots, \psi_n(\cdot;\lambda)$ of functions that solve the Helmholtz equation $\Delta\psi + \lambda\psi = 0$ in Ω *exactly*. Every member of the span is then an exact solution of the PDE, and only the boundary condition remains to be satisfied. Sampling the boundary at m_B collocation points and the interior at m_I points gives two matrices,

$$(A_B)_{ij} = \psi_j(x_i^B), \qquad (A_I)_{ij} = \psi_j(x_i^I),$$

and the question "is λ an eigenvalue?" becomes "is there a coefficient vector making the boundary trace small without making the function small everywhere?" That is measured by the *tension*

$$\sigma_1(\lambda) \;=\; \min_{c \neq 0} \; \frac{\|A_B(\lambda)\, c\|}{\|A_I(\lambda)\, c\|}, \tag{1}$$

the smallest generalized singular value of the pencil $\big(A_B(\lambda), A_I(\lambda)\big)$. The interior block is what stops the minimizer from returning the zero function; it is a normalization, not a quadrature, and its only job is to keep $\|A_I c\|$ away from zero.

Three properties of σ_1 shape everything below.

1. **It dips to a floor at each eigenvalue and is $O(1)$ elsewhere.** The elevated region between dips is the *plateau*, and its height is the natural scale against which smallness is judged.

2. **One evaluation returns the whole ascending vector** $\sigma_1 \le \sigma_2 \le \cdots$. An eigenvalue of multiplicity m drives m of them to the floor together, and reading σ_2 costs nothing extra. Section 6 spends this.

3. **Evaluations are expensive.** One costs a QR, an SVD and a generalized SVD — 5–70 ms depending on basis size. Cost is therefore counted in evaluations and in nothing else, and every design decision below is ultimately a decision about how many to spend.

1.2 The output contract, and its two metrics

Definition 1 (The bracketing problem)**.** Given the tension function, a window $[a, b]$ and a requested relative precision prec, return a set of non-overlapping intervals, each of which the method *believes* contains an eigenvalue.

Two properties of the returned set are hard requirements.

Coverage. Every eigenvalue in $[a, b]$ lies in some returned interval. A lost eigenvalue cannot be recovered downstream by any amount of refinement, so this is the metric that governs.

Purity. No returned interval contains eigenvalues spanning more than prec $\cdot\, \lambda$. An interval holding two eigenvalues the user asked to have separated is a wrong answer, not an imprecise one.

Everything else — how many intervals, how many of them are empty, what it all cost — is reported and traded rather than required.

Remark 1 (The stage must commit). The returned intervals are the ones the method believes hold an eigenvalue, *not* a subdivision of $[a, b]$. The distinction is not pedantic: a method that partitions the window achieves perfect coverage for free and without evaluating anything, while a method that commits covers a small fraction of the window and has to earn it. Without this rule the two are not comparable at all.

Note what purity does *not* constrain: the *width* of an interval holding a single eigenvalue. Such an interval spans zero eigenvalues however wide it is. This observation is worth remembering; it is the whole of §7.2's argument for stopping early.

2 The local model, and what its two parameters mean

Near a simple eigenvalue λ^*, the tension is well described by

$$\sigma_1(\lambda) \;\approx\; \sqrt{\varepsilon^2 + c^2\,(\lambda - \lambda^*)^2}, \tag{2}$$

a cone of slope c with a rounded bottom of depth ε. On the four domains where it has been checked — rectangle, disk, equilateral triangle, L-shape — (2) fits the far field to a median relative residual of 1–3%, failing only where the floor comes close to the plateau, that is, where the basis cannot represent the eigenfunction at all.

The two parameters are independent, and *which is which* matters more than either value.

ε **is the floor.** It is σ_1 at the eigenvalue: the residual of the best eigenfunction the finite basis can represent. It is dimensionless, and it is a property of **the basis**.

c **is the wall slope.** It is $d\sigma_1/d\lambda$ away from the eigenvalue, it carries units, and it is a property of **the domain**.

Figure 1 is the evidence. On the L-shape with a Fourier–Bessel basis at the reentrant corner, taking 20, 40 and 60 terms moves the floor from 4.0×10^{-5} to 1.1×10^{-12} — seven orders — while the wall slope moves from 1.6022 to 1.5951, four parts in a thousand. Enriching the basis deepens the well; it does not change its walls.

2.1 The resolution limit

Setting $\lambda - \lambda^* = \varepsilon/c$ in (2) gives $\sigma_1 = \sqrt{2}\,\varepsilon$: the ratio

$$\frac{\varepsilon}{c} \tag{3}$$

is the scale at which the cone stops looking like a wall and starts looking like its own rounded bottom. Two eigenvalues closer together than ε/c have overlapping bottoms and cannot be separated by *any* method reading this curve, however many evaluations it spends. Since c is fixed by the geometry, ε alone sets the limit — which is the concrete form of the statement that *the search is bounded by the basis*. It is the basis planner's job to make ε small enough; it is the search's job to find everything that is separated by more than about $20\,\varepsilon/c$, which is the quantitative form of the contract between the two layers.

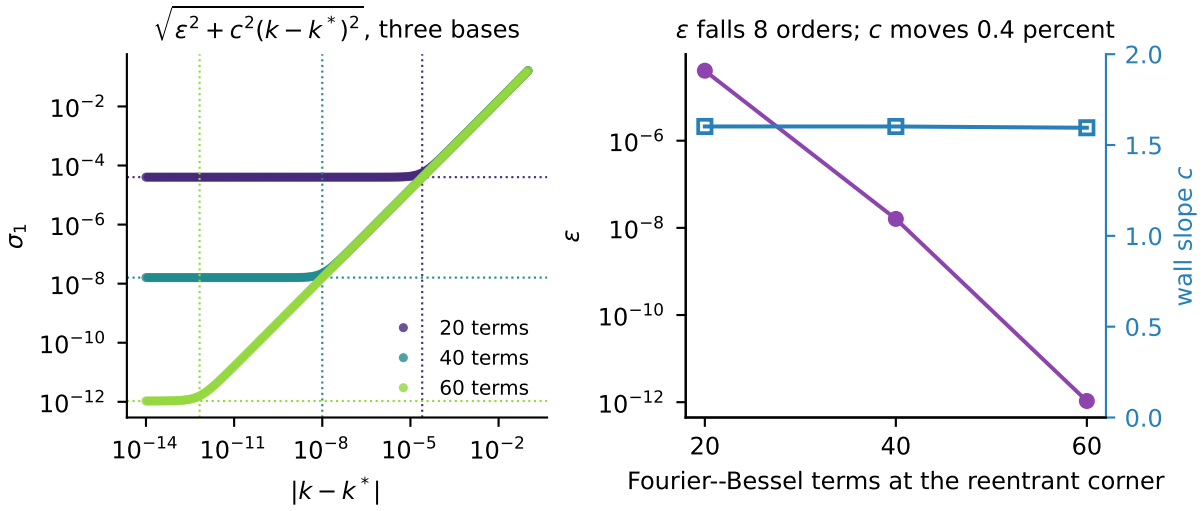


Figure 1: The cone model on the L-shape, at the first eigenvalue, with a Fourier–Bessel basis at the reentrant corner. *Left:* measured σ_1 (points) against (2) (lines) for three basis sizes; dotted lines mark ε and ε/c. The three curves share a wall and differ only in where they bottom out. *Right:* the same three fits. Only ε responds to the basis.

Remark 2 (No absolute threshold on σ can serve as a test). The floor is not a single number even within one scan, and the reason is worth stating precisely because a loose version of it is easy to repeat. At a corner of interior angle ω the local expansion of an eigenfunction runs in

$$r^{\,l\pi/\omega}\,\sin\!\big(l\pi\theta/\omega\big), \qquad l = 1, 2, \ldots,$$

whose exponents are fractional unless ω/π divides l. Every Dirichlet eigenfunction *vanishes* at the corner — that is the boundary condition, and it distinguishes nothing. What distinguishes one mode from another is whether the *coefficients of the fractional-exponent terms* vanish. A mode with no singular part is analytic there and a finite basis represents it almost exactly; its neighbours carry the singularity and are represented far less well. On the L-shape, $\omega = 3\pi/2$ gives exponents $r^{2l/3}$, smooth only when $3 \mid l$; the mode at $\lambda = 2\pi^2$ is the unit square's fundamental $\sin(\pi x)\sin(\pi y)$ tiled across the three cells, which vanishes on the whole integer grid and is entire.

Every threshold in the algorithm is therefore stated as a fraction of the *measured plateau*, never as an absolute tension.

The measurement behind this rule has itself been corrected, and the correction is instructive. An early measurement recorded a five-order floor spread on the L-shape — 3.1×10^{-13} at one level against 5.5×10^{-8} at a neighbour — and explained it structurally. With a recipe-grade basis (80 pure Fourier–Bessel terms at the reentrant corner) the same levels read within a factor of 4.2 of one another, and the level previously at 5.5×10^{-8} reads 5.3×10^{-15}. So the five orders were an artifact of an inadequate basis. The *design rule* survives either way, because a factor of four still defeats any fixed constant and a domain whose basis cannot be made adequate will still show the large spread; what does not survive is the claim that the L-shape is intrinsically such a domain.

3 Why the wavenumber, and why uniquely

This section contains the shortest derivation in the document and the one that decides the most.

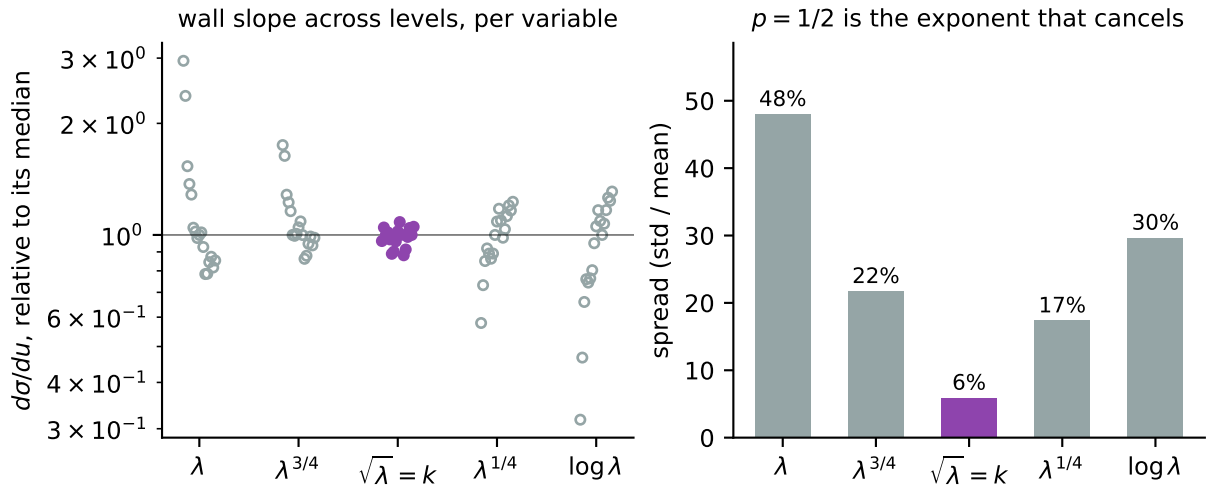


Figure 2: The wall slope across 17 isolated levels of a $1 \times \varphi^{-1}$ rectangle, under five changes of variable. *Left:* each level's slope relative to the median for that variable. *Right:* the spread. The exponent that cancels is the one the data pick out.

The cover phase of §5 rests on a single Lipschitz bound L used across a whole window. Such a bound is conservative by the ratio of the largest local slope in the window to the smallest, and it is paid *everywhere*. Reducing that ratio is therefore the largest available saving, and it is available before any algorithm has been chosen.

Measurement supplies the scaling: on the family where it was first taken,

$$c(\lambda) \;=\; \frac{C}{\sqrt{\lambda}}, \qquad C = 0.27 \pm 8\%, \tag{4}$$

so c varies by a factor of 4.81 across $\lambda \in [10, 400]$. Now search not in λ but in $u = \lambda^p$ for some power p. By the chain rule,

$$\frac{d\sigma}{du} \;=\; c(\lambda)\,\frac{d\lambda}{du} \;=\; \frac{C}{\sqrt{\lambda}} \cdot \frac{\lambda^{1-p}}{p} \;=\; \frac{C}{p}\,\lambda^{1/2-p}, \tag{5}$$

which is constant in λ for $p = 1/2$ and for no other power. This is not a fit; it is the exponent that cancels.

Proposition 1. *Under the scaling (4), $u = \lambda^p$ makes the wall slope independent of the level if and only if $p = 1/2$; that is, if and only if u is the Helmholtz wavenumber $k = \sqrt{\lambda}$.*

That k is also the physical variable — the wavenumber at which the basis functions are built — is the reason this is expected to transfer to other domains, rather than being a property of the family it was measured on.

Figure 2 confirms it on a different rectangle, in a different implementation, from the one where (4) was measured. Over 17 isolated levels in $\lambda \in [10, 520]$, the spread of $d\sigma/du$ (standard deviation over mean) is 48% in λ, 22% in $\lambda^{3/4}$, 17% in $\lambda^{1/4}$, 30% in $\log \lambda$, and **5.9% in** k. The originating measurement reported 57%, 27%, 17%, 32% and 5%. Two independent implementations, two different rectangles, the same answer.

Remark 3 (How much this is worth). In the design's own version history this change of variable accounts for a factor of 1.8 out of a total 2.6 improvement — more than every algorithmic change in the method combined. It is worth stating plainly, because it is easy to read as a numerical detail and it is not one.

3.1 Two consequences of working in a transformed variable

The precision factor. Purity is judged on $d\lambda$ against $\mathrm{prec}\cdot\lambda$, but the search contracts brackets in u. A bracket $[u_1, u_2]$ maps to $d\lambda = (d\lambda/du)\,du$, so for $u = \lambda^p$ the relative precision that must be met in u is

$$\mathrm{prec}_u \;=\; p\cdot\mathrm{prec}, \tag{6}$$

a *half* for the wavenumber. Evaluated at the upper end of the bracket, so conservative.

The ulp slack. The round trip $k \mapsto k^2$ is not exact. A bracket that correctly fences k^* can fail to fence $\lambda^* = (k^*)^2$ after squaring, because an endpoint rounds the wrong way; this was observed as a coverage loss of exactly one unit in the last place. Every endpoint is therefore widened outward by four ulps of λ when the bracket is converted (`Objective.in_lam_bracket`, constant `BRACKET_ULPS`). The same constant is the refiner's margin for "the same point", and the two interact; §9.5 records how.

And the refiner wants the other variable. §9 shows that the quadratic fit the refinement performs is markedly better in λ than in k. The two stages want two variables and they get them: **cover in k, refine in λ.** This is not a compromise between two measurements but the conclusion of both, and it is worth noticing that the reasons are genuinely different — the scan cares about the *slope*, which is flat in k, and the refiner cares about the *curvature*, which is better behaved in λ.

4 Seeding the scan, and estimating the slope

4.1 Weyl sizes the grid and says nothing else

Weyl's law with the perimeter correction,

$$N(\lambda) \;\sim\; \frac{|\Omega|\;\lambda}{4\pi} \;-\; \frac{|\partial\Omega|\,\sqrt{\lambda}}{4\pi}, \tag{7}$$

predicts the number of eigenvalues in a window to 1–11% on the families tested. It is used for exactly one thing: to set the starting sample density, at 1.5 samples per expected gap. It says nothing whatever about *where* anything is, and must never be used as though it did.

The starting density barely matters, because the doubling below adapts: 0.5, 1.0 and 1.5 samples per gap all cost the same. The larger is chosen for margin.

4.2 Nested doubling, and why $n \to 2n$ is wrong

The seed then doubles, inserting midpoints so that $n \to 2n-1$ and each grid *contains* the previous one exactly, until the largest observed secant slope stops growing by more than 10%.

Exact nesting is not tidiness. Under $n \to 2n$ the new points land at irrational offsets from the old ones, and some of them land within an ulp of an old point. A secant computed across such a

pair measures rounding, not slope: one such estimate reached 7.3×10^{10} against a true slope of 0.6. Since the slope model is monotone non-decreasing (§4.4), a model that once believes that can never recover. For the same reason, pairs closer than 10^{-9} of the sampled span are excluded from the secant estimate altogether. Midpoint insertion has a second benefit: the old points are reused bit-identically, so a doubling costs only the genuinely new samples.

4.3 The bootstrap hazard runs opposite to intuition

It is natural to worry that an *over*estimated Lipschitz constant will make the search expensive. The dangerous error is the other one. As §5 will show, a sample of value v excludes a neighbourhood of radius v/L; an *under*estimated L makes those radii too **large**, so a search that has not yet seen the true slope excludes everything and terminates having found nothing — silently, with no error at all. Measured, the largest observed secant as a function of seed density, against a true slope of 0.0656:

n	8	16	32	64	128	256
largest secant	0.0046	0.0128	0.0386	0.0599	0.0647	0.0652
as a fraction of the truth	7%	20%	59%	91%	99%	99%

At $n = 8$ the exclusion radii would be fourteen times too large.

Two things defuse this. First, the doubling itself: the estimate is not used until it has stopped growing. Second, and more important, *exclusions are recomputed from the current L on every pass*, so a subsequent growth in L revives whatever that growth un-excludes. An underestimate therefore costs extra work rather than causing a silent miss.

Remark 4 (The bootstrap is the falsification test). An estimate that never settles is not a nuisance to be worked around: it is the evidence that nothing bounds the local slope on this curve, at any scale sampled. That is a legitimate ground for the method to decline (§10.3), and it is the only such evidence available.

4.4 The slope model, and what it is honestly worth

The model is

$$L \;=\; s \cdot \max_i \left| \frac{v_{i+1} - v_i}{x_{i+1} - x_i} \right|, \qquad s = 1.5, \tag{8}$$

monotone non-decreasing over the run. The honesty clause must be stated before anything is built on it.

Proposition 2 (There is no certificate here)**.** *No finite sample set bounds a Lipschitz constant from above. The largest observed secant is a* lower *bound on* $\sup |f'|$*, and multiplying it by a safety factor produces a number with no formal status.*

This is not a defect that better engineering removes; it is a property of the information available. The consequence is that every completeness claim the method makes is *conditional on* L, and must be reported as such (§10).

Two measurements set $s = 1.5$, and they pull in opposite directions.

- **The bound is nearly tight, so it cannot be shaved.** On the window where it was measured the largest observed secant reaches the true nearest eigenvalue exactly. Using the 99th-percentile slope instead of the maximum — a 1.4% reduction — wrongly excluded 13 of 600 samples. Coverage is lost by very small underestimates.

- **Margin is cheap in the seed.** Raising s from 1.2 to 2.0 cost 40% more evaluations for identical coverage.

Finally, distinguish a *surprise* growth — L rising after exclusions have already relied on it — from mere convergence of the estimate. A creep from 0.0647 to 0.0656 is the estimate settling. Only surprises are counted, and a run with many of them deserves a rerun at a larger s; it is not, by itself, wrong.

4.5 An aside: could the scan have found it at all?

Sizing the scan is a *precondition*, not a diagnostic, and `chladni` ships a closed-form check for it. Model the dip as a clipped V, $\sigma \approx \min(\sigma_{\text{base}}, C_k\,|k - k_j|)$, where σ_{base} is the plateau. Its visible width is $w_k = 2q\,\sigma_{\text{base}}/C_k$ for a visibility fraction q, and demanding that a scan of stride Δk resolve it, $w_k \geq \Delta k$, gives

$$\sigma_{\text{base}} \;\geq\; \frac{\kappa}{2q} \cdot \frac{|\Omega|}{|\partial\Omega|} \cdot \Delta k, \qquad \kappa \;=\; \frac{C_k}{|\Omega|\,/\,|\partial\Omega|}. \tag{9}$$

The $|\Omega|\,/\,|\partial\Omega|$ scaling is confirmed exactly: the same $3:1$ rectangle at scales 0.5, 1, 2 gives $C_k/(A/P) = 2.4600,\ 2.4600,\ 2.4598$, while C_k/A varies by a factor of four over the same three. The dimensionless κ transferred between two independent implementations within one standard deviation, 2.591 ± 0.226 against 2.384 ± 0.231.

5 Cover by exclusion

5.1 The Piyavskii bound

Suppose f is L-Lipschitz and we know $f(x_i) = v_i$ and $f(x_j) = v_j$ at two adjacent samples, $x_i < x_j$, $h = x_j - x_i$. Then for every $x \in [x_i, x_j]$,

$$f(x) \;\geq\; \max\big\{\, v_i - L(x - x_i),\;\; v_j - L(x_j - x) \,\big\}, \tag{10}$$

the pointwise maximum of two cones of slope L dropping from the two samples. The right-hand side is minimized where the two branches cross, giving

Lemma 1 (The envelope and its minimizer)**.** *The bound* (10) *attains its minimum*

$$\hat{s} \;=\; \frac{v_i + v_j - Lh}{2} \qquad \textit{at} \qquad \hat{x} \;=\; \frac{x_i + x_j}{2} + \frac{v_i - v_j}{2L}. \tag{11}$$

Proof. The two branches are affine with slopes $-L$ and $+L$; their maximum is minimized at their crossing, $v_i - L(x - x_i) = v_j - L(x_j - x)$, which rearranges to $\hat{x}$ above and gives the value $\hat{s}$ by substitution. □

Two consequences, and the method is built from them.

The exclusion certificate. If $\hat{s} \geq 0$ then $f > 0$ throughout (x_i, x_j), so no eigenvalue lies between the two samples. The cell is *certified empty* — conditional on L, per Proposition 2.

Where to look next. $\hat{x}$ is the point attaining the bound: the most informative place in the cell, and where the search samples.

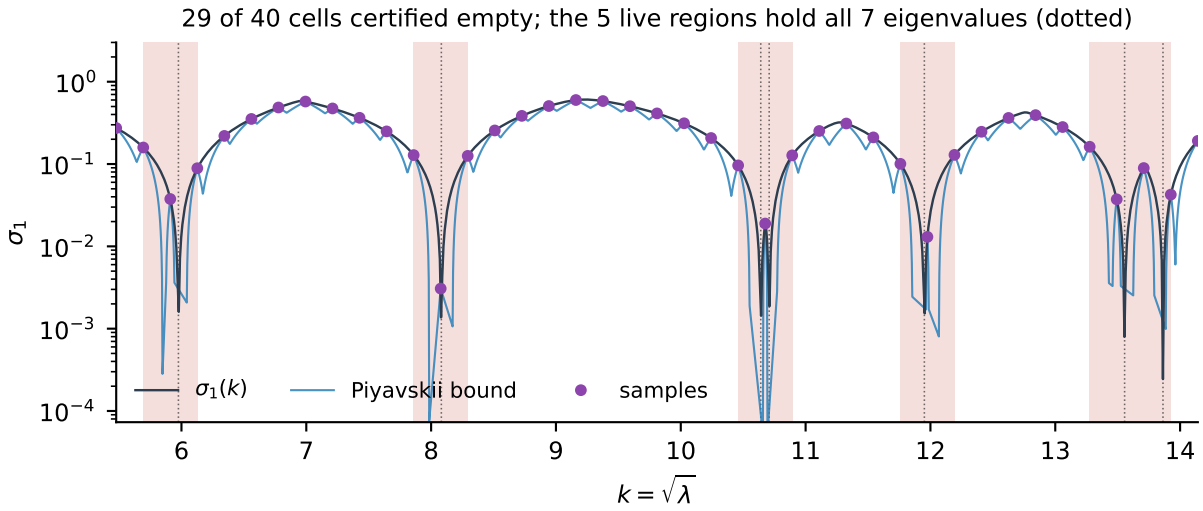


Figure 3: Cover by exclusion on a $1 \times \varphi^{-1}$ rectangle, $\lambda \in [30, 200]$, from 41 uniform samples with $L = 1.5\times$ the largest observed secant. The sawtooth is the Piyavskii bound (10), drawn only where it is positive. 29 of 40 cells are certified empty; the five surviving live regions (shaded) contain all seven eigenvalues (dotted). Two of them hold a pair, which is what §6 exists to detect.

Cells not excluded are *live*, and contiguous live cells are merged into *regions*. Merging matters: the two cells either side of an eigenvalue are both live, and reported separately they would look like two candidates where there is one.

Figure 3 shows the construction on a real tension curve.

Remark 5 (The zero threshold, and why it does not need to be raised). The test $\hat{s} \geq 0$ presumes σ_1 reaches zero at an eigenvalue, and it does not: it bottoms at $\varepsilon > 0$. So as the spacing shrinks the bound rises toward a positive value and a sufficiently finely sampled well could, in principle, exclude *itself*. This is a reasonable objection and the answer is structural rather than a tuned constant: the cover loop stops on purity-by-width long before that regime is reached, so the zero threshold never bites. Replacing it with a positive threshold was tried and broke every case — 0 of 66.

5.2 The cost law

What does it cost to certify a cone's wall as empty? Take two samples on the same wall at distances $u_a > u_b$ from the eigenvalue, so their values are $c\,u_a$ and $c\,u_b$, and take $L = s\,c$. By (11) the cell between them is excluded when

$$\frac{c(u_a + u_b)}{2} - \frac{s\,c\,(u_a - u_b)}{2} \;\geq\; 0 \quad\Longleftrightarrow\quad (1 - s)\,u_a + (1 + s)\,u_b \;\geq\; 0,$$

that is, when

$$\frac{u_b}{u_a} \;\geq\; \frac{s - 1}{s + 1}. \tag{12}$$

Proposition 3 (The ladder). *The condition* (12) *is independent of* c. *A cone's wall is therefore covered by a* geometric ladder *of samples with ratio at most*

$$r(s) \;=\; \frac{s + 1}{s - 1},$$

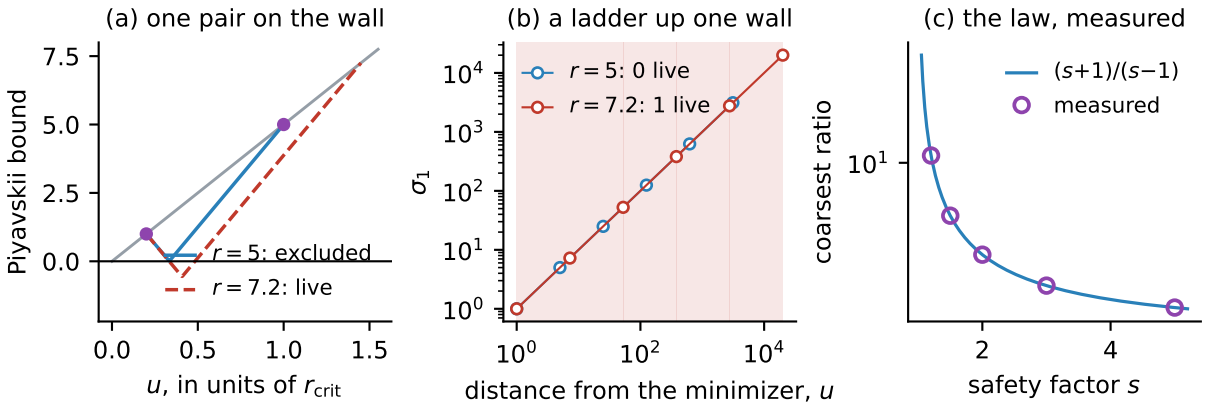


Figure 4: *(a)* One pair on a wall. At the critical ratio the Piyavskii bound just touches zero; past it the bound dips below and the cell stays live. *(b)* A ladder up one wall: at $r = 5$ every cell is excluded, at $r = 7.2$ none is. *(c)* The ratio measured by bisection against `live_regions`, against $(s+1)/(s-1)$.

and covering a range of scales R costs $\log R / \log r(s)$ samples per side.

Two things about this deserve emphasis. It is independent of the wall slope, so it holds uniformly over the window and over domains. And it is exact: Figure 4 recovers $r(s)$ to five figures by bisecting on the ratio against the exclusion code itself.

Remark 6 (A better prediction cannot help the cover phase). Proposition 3 retires a family of ideas. Placing a sample *exactly on* an eigenvalue leaves the live region completely unchanged, and so does placing two that straddle it to within 10^{-9}: what is live is the *wall*, and the wall yields to the ladder or to nothing. Six separate attempts to make cover cheaper by sampling more cleverly all failed for this one reason, and a seventh would too. Cleverness about *where the eigenvalue is* belongs in the locator (§7), not here.

Remark 7 (A larger safety factor is a finer ladder). Raising s does not merely buy margin; it costs $1/\log r(s)$ per side, so the cover phase's cost grows as s rises. That is why the tension in §4.4 between "the bound is nearly tight" and "margin is cheap" is a real trade and not a free choice.

5.3 Why adapt at all

A uniform grid must resolve the tightest pair it might meet. A pair at separation d is resolved once the sampling near it is finer than about $d/3$, so a uniform grid over $[a, b]$ needs $n \sim 3(b-a)/d$ samples *everywhere*, while exclusion refines only where σ_1 is small and costs $O(\log(1/d))$. Measured on an interval of width 2:

d	uniform grid	exclusion	ratio
3×10^{-3}	2048	47	43×
10^{-4}	$\sim 6 \times 10^4$	57	1053×
10^{-6}	$\sim 6 \times 10^6$	71	8.5×10^4
10^{-10}	$\sim 6 \times 10^{10}$	97	6×10^8

The comparison matters because d is *not known in advance*. A black-box solver cannot tune its grid to the tightest pair it will meet, so the uniform-grid column is a cost it would have to pay everywhere and forever, against a pair it may or may not have.

6 The isolation gate: what σ_2 buys

6.1 Why a gate is needed

Once a region is believed to hold one eigenvalue, a fast locator can drive it down cheaply. But a locator that converges to a single minimizer will happily converge to *one member of a close pair* and leave the other outside the bracket it returns. This is not hypothetical: it is exactly how a grid-then-bisect baseline falls from coverage 1.0 to 0.577. The fast path must therefore be gated, and the gate must be right.

6.2 The signal, and why it is conservative by construction

One evaluation returns the whole ascending vector, so σ_2 is free. Near an eigenvalue, σ_2 is the second branch of the pencil — which, at a simple level, is the *next* eigenvalue's cone — so

$$\frac{\sigma_2(x)}{c} \;\approx\; \text{distance from } x \text{ to the next eigenvalue.} \tag{13}$$

A region is certified to hold one eigenvalue when

$$\text{neighbour} \;>\; \gamma \cdot (\text{region width}), \qquad \gamma = 1.5. \tag{14}$$

The estimator's error structure is what makes this safe, and it is worth stating carefully. Measured against the true gap, (13) reads at ratio 0.97–1.03 when the neighbour is close, and 0.78–1.30 when it is far (Figure 5). It is accurate *precisely where the decision is hard* — when a neighbour is close enough to be a purity risk. Where it is inaccurate it errs in both directions, but only for neighbours many region widths away, where a 30% misreading does not change the answer to "is the neighbour further away than this region is wide?". Close in, where a misreading could flip that answer, the meter is good to 3%, and the slope it divides by is an upper bound (rule 2 below), which biases the reading toward refusing to certify.

6.3 Estimating c without cancellation

The gate needs c, and it reads it off the descent triple. If sampled points $\mathrm{lo} < \mathrm{mid} < \mathrm{hi}$ straddle the minimizer then, on a cone, $v_{\mathrm{lo}} = c(\lambda^* - \mathrm{lo})$ and $v_{\mathrm{hi}} = c(\mathrm{hi} - \lambda^*)$, so adding,

$$c \;=\; \frac{v_{\mathrm{lo}} + v_{\mathrm{hi}}}{\mathrm{hi} - \mathrm{lo}}. \tag{15}$$

No subtraction of nearly equal quantities appears anywhere, and the estimate does not depend on where λ^* is within the triple.

6.4 Four rules, each of which cost a measured failure

1. **Judge the region, not the triple inside it.** The region is what gets handed to the locator and can be several times wider than the certificate triple. A pair 8.4×10^{-4} apart was certified inside a region 1.24×10^{-3} wide, and then collapsed onto one member.

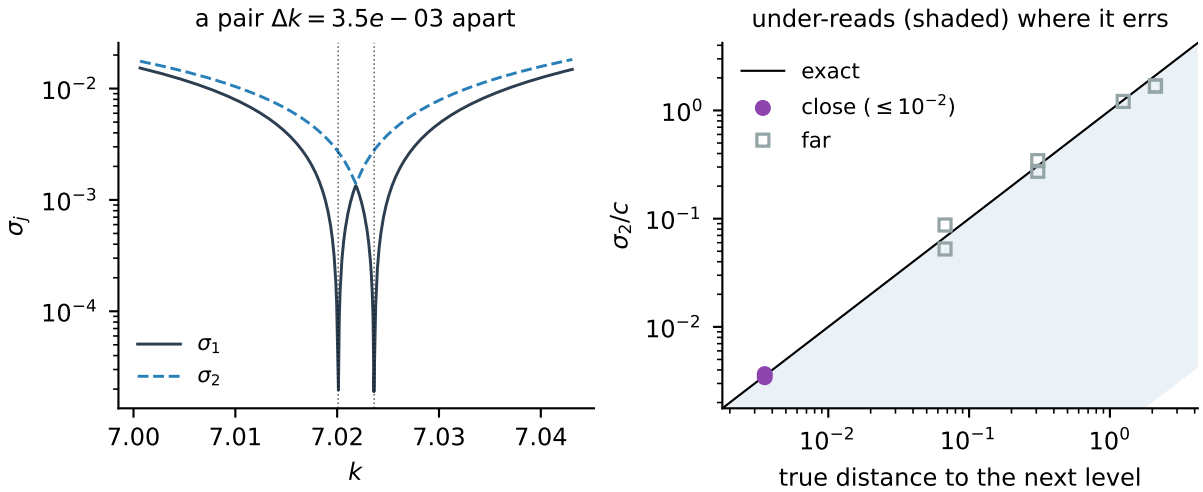


Figure 5: *Left:* σ_1 and σ_2 across a resolved pair on a $(1+\delta)\times 1$ rectangle. σ_2 has a single minimum, at the crossing point. *Right:* the reading σ_2/c against the true distance to the next level, over nine levels of two pencils, computed by the searcher's own `isolation` routine. The shaded half-plane is where the meter under-reads.

2. c **must be an independent *upper* bound**, taken as $\max(c_{\text{local}}, L)$. Where a region holds *two* eigenvalues its flanks span both cones, so hi − lo in (15) is too wide, c comes out too small, the neighbour appears further away than it is, and the gate certifies precisely what it must not. Trusting the local estimate alone cost 3 of 26 eigenvalues.

   This is worth dwelling on, because the instinct is the reverse. Replacing the crude global L with a *learned* wall slope — an obviously better estimate — loses 12 of 19 cases. A smaller slope makes the estimated neighbour distance larger, which pushes the gate toward certifying. The global bound's apparent excess conservatism *is* the safety margin.

3. **Silence is not permission.** If no descent certificate exists in the region there is no decision to be had, and the region stays on the slow path. "I cannot tell" must not be recorded as "yes".

4. **Do not read the level's multiplicity off the σ vector here.** Counting branches at the floor and reading the first one that is not is correct *in principle* — at a multiplicity-2 level, σ_2 is that level's own second branch, and the next distinct level is reported by σ_3 — and it saves 40% on exactly degenerate cases. It is also ambiguous at range by construction: approaching an unresolved pair of separation d from distance u,

$$\frac{\sigma_2}{\sigma_1} \;=\; 1+\frac{d}{u}, \tag{16}$$

   which is 1.0006 at the start of a cover pass. No threshold separates that from a genuine multiplicity-2 level. Swept over two windows, the failure count is **not monotone** in the threshold — $0,1,0,1,2$ failures at $1.0, 1.05, 1.2, 1.5, 2.0$ — and a parameter whose safety alternates as it is tightened has no safe setting, only settings that pass the cases tried.

Remark 8 (The reading is not wrong; its placement was). Read *after* location, against a deliberately wide bracket, the same quantity is unambiguous: 14 levels, 14 correct multiplicity readings, at

zero additional evaluations, because every point needed is already in the cache. Within a multiplet the ratio σ_{j+1}/σ_j runs 1.0–1.7 and across the gap 10^{11}–10^{15}, so a threshold of 10 is not a delicate parameter at all. What (16) forbids is reading it *at range*, which is the only place the bracketer could.

7 Locate: predict, then straddle

7.1 The prediction is exact, and it is not enough

Under the cone model (2),

$$\sigma_1^2 \;=\; \varepsilon^2 + c^2(\lambda - \lambda^*)^2 \tag{17}$$

is *exactly* quadratic. Three points therefore determine the minimizer exactly rather than approximately, and the residual error comes only from the two branches meeting at the bottom having slightly different slopes. This is also far better conditioned than the obvious alternative of extrapolating a wall to zero, which subtracts nearly equal values.

Squaring is not a numerical sin here; see Remark 13.

Now the insight that makes the locator work.

Remark 9 (Prediction alone does not tighten a bracket). A bracket needs sampled points on *both* sides of the minimizer, so its width is set by whichever side of the prediction has no sample near it. The prediction error has a *systematic sign*, because the two branches have slightly different slopes. So every scheme that samples *at* the prediction converges from one side only, and the far flank never moves.

The fix is to place the sample a fixed fraction of the bracket to one side of the prediction, on whichever side is not already fenced. Each round of two such samples shrinks the bracket to about $2\times$ the straddle fraction. That fraction is bounded below by the prediction error — an offset smaller than the error lands on the wrong side and the triple stops being a descent — and above by the contraction it buys.

Figure 6 measures both bounds. The parabola's vertex lands within 9.4×10^{-5} of the truth at the median and 3.1×10^{-3} at the 90th percentile, as fractions of the bracket; the cost rises steadily with the offset above that. The shipped value 0.02 sits about six times above the measured 90th percentile of the prediction error, and costs 14 evaluations against 12 at 0.01 on this pencil.

Remark 10 (Three designs that fail, each plausible). Plain interpolation — sampling at the prediction — stalled with the bracket stuck. Bisecting the lopsided side cured the stall but cost no less than exclusion. Mirroring the near sample across the prediction oscillated without converging. Each is a natural thing to try, and each is worth knowing has been tried.

7.2 What licenses emitting a bracket

Every endpoint must be sampled, and carry a descent certificate. For three sampled points $p < q < r$ with $f(p) > f(q) < f(r)$, continuity alone gives a local minimum in (p, r). Nothing weaker may define an endpoint. In particular an *extrapolated* root may not: two-point secant roots fitted from opposite walls of a close pair were measured to **cross** — the left root landing to the right of the right root — yielding an inverted bracket, or a sliver containing nothing. Extrapolation may choose *where to sample*; it may not name an endpoint.

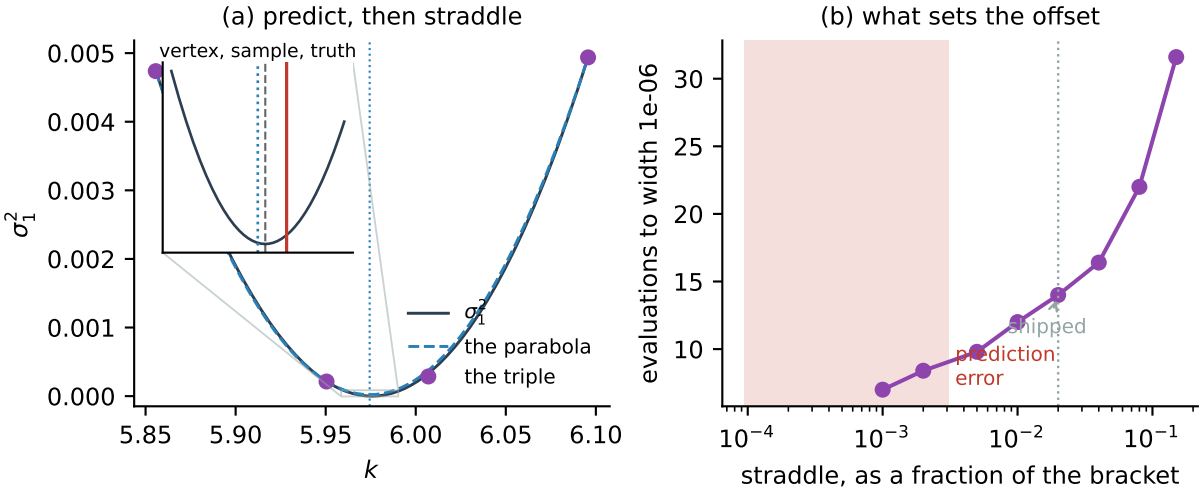


Figure 6: *(a)* One locator step. σ_1^2 and the parabola through the triple are indistinguishable at full scale; the inset zooms on the bottom, where the predicted vertex (dotted), the sample placed 2% to its unfenced side (solid) and the true minimizer (dashed) separate. *(b)* Evaluations to contract a bracket to width 10^{-6}, averaged over five isolated levels, against the straddle fraction. The shaded band is the measured prediction error over the same runs, median to 90th percentile: below it the offset is inside the noise of the prediction.

Two details of the certificate are easy to get wrong, and were.

- **Anchor on the region's lowest sample**, not on the deepest-looking triple. This is what makes the bracket tighten monotonically: every new sample nearer the bottom becomes the new centre, and its immediate neighbours are necessarily closer than the old ones. Selecting by depth instead stalls, because depth tends to 1 as the flanks approach the middle.
- **Immediate neighbours only.** A descent read off non-adjacent samples would skip whatever lies between them, and could straddle two cones with a barrier in between.

Depth, normalized by the further flank. To decide whether a region holds a minimizer at all, the middle's value is compared to its flanks. The normalization must be by the *larger* of the two flanking values. Using the nearer one systematically rejects real eigenvalues: when a cone bottoms out near the middle of a sampling cell its two flanking samples are almost equidistant, so their values are almost equal and the ratio approaches 1 however deep the cone actually is. That cost 27% of coverage — 19 of 26 eigenvalues instead of 26 — *at every grid density*, because refining the grid only moves the problem to a different cell. Against the further flank, a mid-cell cone scores about 0.33 while plateau ripple still scores about 0.98, which is the separation the ratio is for. The test is asked *once* per region, and never to choose among certificates: depth necessarily tends to 1 as a bracket tightens, so a locator that filtered on it would stall partway down.

7.3 Stop at isolation

Recall from §1 that purity does not constrain the width of a bracket holding one eigenvalue. Refining past isolation therefore buys nothing the contract asks for, and the refiner of §9 reaches

machine precision from a wide bracket in about seven evaluations. So the bracket is handed on once the neighbour is four bracket-widths clear.

State plainly what this gives up. A bracket narrower than $\mathrm{prec} \cdot \lambda$ cannot be impure *however wrong the gate was*; that unconditional guarantee is traded away, and purity then rests on §6. This is the change in the design most in need of confirmation on domains beyond the one it was measured on.

Remark 11 (prec is a stopping rule, not a merge licence). The search never reasons about whether merging two nearby eigenvalues into one bracket is *permitted*. The economics are entirely one-sided: merging saves the refinement of one bracket, certifying that a merge is legal means resolving both eigenvalues anyway, and being wrong is a hard purity failure. So the merges the search performs are only the ones it could not resolve, and splitting beyond the licence is explicitly not an error.

8 The algorithm assembled

Every ingredient is now defined, and it is worth seeing them composed before the refinement stage is added, because two structural facts are visible in the control flow and invisible in the prose.

One loop, not two. The obvious structure — cover the window to find candidates, then refine each candidate separately — is wrong, and the baselines measure how wrong. Any per-candidate refiner anchored on "the lowest point I can see" converges on one member of a close pair and leaves the other outside the interval it returns; this is exactly how grid-then-bisect falls from coverage 1.0 to 0.577. So cover and locate *alternate*, and they operate on *regions* rather than on candidate points. A region holding two eigenvalues then splits of its own accord, the moment a sample placed between them is excluded.

The gate is asked three times. It is consulted to decide whether a region may leave the cover loop (Algorithm 2, line 3), again after *every* locator step (Algorithm 3, line 9), and once more on the way out (line 14). Each catches a different failure, and all three are free: the gate is a pure function of samples already paid for.

That last point is worth stating as a rule, because it explains the shape of the whole algorithm.

Remark 12 (Only one line spends anything). In all three routines the sole operation that costs an evaluation is Evaluate. LiveRegions, DescentCertificate, Gate, LocatorStep and LooksLikeADip are pure functions of the sample arrays. This is why the design can afford to re-ask expensive-sounding questions at every step, and why the cost accounting in §10 counts cache *misses* and nothing else.

Algorithm 1: BRACKET — the top level

input : tension oracle $f(x) \mapsto (\sigma_1, \sigma_2, \dots)$; window $[a, b]$; precision prec; flag *rescue* (off by default; §8.2)
output : brackets in λ, with diagnostics — or a decline

work in $k = \sqrt{\lambda}$; $\text{prec}_u \leftarrow \text{prec}/2$; // (6)
$(X, V, \text{settled}) \leftarrow \text{SEED}(f, a, b)$; // Weyl density, then nested doubling, §4
if $\neg$ *settled* **then return** DECLINE("nothing bounds the local slope");
$S \leftarrow \text{median}(V)$; // the plateau, measured once; Rem. 2
$L \leftarrow s \cdot \max_i |(V_{i+1} - V_i)/(X_{i+1} - X_i)|$; // (8), $s = 1.5$
budget $\leftarrow 600 \cdot \max\big(N(b) - N(a),\, 1\big)$; // per eigenvalue, never flat; without a Weyl count, N is the number of live regions
floorLimited $\leftarrow$ FALSE;
for *round* $\leftarrow 1$ **to** 5 **do**
  $(X, V, L, \mathcal{R}, \text{stalled}) \leftarrow \text{COVER}(\dots)$; // Alg. 2
  floorLimited $\leftarrow$ floorLimited $\vee$ stalled;
  $(X, V, \mathcal{R}, \text{reopened}) \leftarrow \text{LOCATE}(\dots)$; // Alg. 3
  if *reopened* $= 0$ **then break**;
if *rescue* **then**
  $\mathcal{R} \leftarrow \mathcal{R} \cup \text{RESCUEEXCLUDEDWELLS}(X, V, \mathcal{R}, S)$; // §8.2
  floorLimited $\leftarrow$ floorLimited $\vee$ (any were added);
$\mathcal{K} \leftarrow \text{CONFIRM}(X, V, \mathcal{R}, L, S)$; // Alg. 3
return each $(a', b') \in \mathcal{K}$, mapped back by $u \mapsto u^2$; each end moves out by 4 ulp of λ when the refiner takes it ; // §3

Algorithm 2: COVER — refine by exclusion until every region is settled

output : samples, a possibly larger L, the live regions, and whether the floor bound

while true **do**
  $\mathcal{R} \leftarrow \text{LIVEREGIONS}(X, V, L)$; // cells with $\hat{s} < 0$, merged; (11)
  $\mathcal{T} \leftarrow \{\, R \in \mathcal{R} \;:\; \neg\,\text{PUREBYWIDTH}(R, \text{prec}_u) \;\wedge\; \neg\,\text{GATE}(R, L) \text{ certifies single} \,\}$; // silence is not permission
  if $\mathcal{T} = \varnothing$ **then return** $\mathcal{R}$, FALSE;
  if *no reduction in* $\sum_{\mathcal{T}} |R|$ *for 200 steps* ***or*** *budget spent* **then**
    return $\mathcal{R}$, TRUE ; // floor-limited, *not* a decline; §10.3
  $i \leftarrow \arg\min\{\, \hat{s}_i \;:\; \text{cells } i \text{ of } \mathcal{T} \,\}$; // the deepest unexplained dip
  $x \leftarrow \hat{x}_i$; EVALUATE f at x and insert into (X, V) ; // the Piyavskii point, (11)
  if x *was already a sample* **then return** $\mathcal{R}$, TRUE;
  if $s \cdot \max \textit{secant} > L$ **then**
    $L \leftarrow s \cdot \max \text{secant}$; growths += 1 ; // exclusions are recomputed next pass
    if *growths* > 16 **then return** DECLINE("the slope estimate keeps growing");

Algorithm 3: LOCATE, then CONFIRM

```
1  procedure LOCATE(X, V, R, prec_u, L):
2      foreach region (a_0, b_0) ∈ R do
3          if PUREBYWIDTH or ¬ GATE certifies single then keep (a_0, b_0); continue;
4          (a, b) ← (a_0, b_0);
5          repeat
6              x ← LOCATORSTEP(X, V, a, b) ;          // vertex of σ_1^2, then straddle; §7
7              if x is NONE or already sampled then break;
8              EVALUATE f at x; update (a, b) to the new certificate;
9              g ← GATE((a, b), L) ;                  // free, and the answer changes
10             if g.neighbour ≤ b_0 − a_0 then break  // a pair after all;
11             if g.single ∧ g.neighbour ≥ 4 (b − a) then break   // isolated; hand it on;
12         until at most 200 times, within budget;
13         g ← GATE((a, b), L);
14         if g is NONE or g.neighbour ≤ b_0 − a_0 then
15             keep (a_0, b_0);   reopened += 1 ;     // too wide beats being wrong
16         else
17             keep (a, b);

18 procedure CONFIRM(X, V, R, L, S):
19     foreach region (a, b) ∈ R do
20         repeat
21             EVALUATE f at the Piyavskii point of the deepest cell in (a, b) ;   // not the
                widest gap
22         until 3 times, or until a certificate exists;
23         keep (a, b) if DESCENTCERTIFICATE exists, or min V ≤ (1/2) max V within it, or
            min V ≤ (1/2) S ;                         // the last disjunct: a region inside a well reads flat
```

8.1 Reading the algorithm

Four lines carry most of the design, and each was arrived at by a measured failure.

- **The TODO filter** (Alg. 2, line 3) has two disjuncts because a region may leave the cover loop for two quite different reasons: it is narrow enough that it *cannot* be impure whatever it holds, or the gate has certified that it holds one eigenvalue. A region the gate cannot judge satisfies neither and stays.
- **The two non-decline exits** in Alg. 2 — stalling, and asking for a point that is already a sample — return the regions in hand with a flag rather than an error. They mean the objective's floor is binding, which is a fact about the pencil and not a failure of the search.
- **The reopen path** (Alg. 3, line 14) discards the locator's work and returns the *original* region. It is what keeps the coverage guarantee when the gate was wrong, and it is why the top level loops at all: the re-opened region goes back through COVER, where the locator's own samples now sit beside one member of the pair and make exclusion cheap.

- **The third disjunct in** CONFIRM exists because a region lying entirely *inside* a well has every sample on the floor, so its internal ratio is ≈ 1 and it reads as flat. Flatness at the floor is the opposite of flatness on the plateau, which is why the test is a fraction of the measured S (Remark 2).

Finally, note what is *not* in the pseudocode: any reasoning about whether two nearby eigenvalues may be merged. The search never decides that. It contracts until a region is pure by width or certified single, and the merges it performs are only the ones it could not resolve.

8.2 Rescuing wells the cover excluded

Remark 5 argued that the zero threshold never bites because the cover loop stops on purity by width first. That argument assumes a floor far below the samples. On an objective whose floor is *raised* — the coarse penalty of §8.3, whose floor sits near 10^{-7} — it fails in one specific case. A simple level is certified single and handed to the locator before its samples get close enough to exclude it. A *double* is never certified single, because its σ_2 sits on the floor too, so the cover keeps sampling into the well until the samples are closer than about $2\varepsilon/L$, and then the envelope test excludes the well's own bottom. Every double of two `douse` polygons (an octagon and an 8-gon) was lost this way.

The remedy is a pass after the cover/locate rounds, enabled by the flag `rescue_floor` of `bracket_interval` and off on the default path. Every sampled local minimum below 10^{-2} of the plateau S (`RESCUE_DEPTH`) that lies outside every live region becomes a region again. The new region runs out to the first sample on each side where σ_1 exceeds ten times the minimum (`RESCUE_RISE`), or to a sampled local maximum if that comes first, and overlapping regions are merged. The hump stop matters: walking past one merged a `douse` pair 1.2×10^{-5} apart into a single bracket, and the count then read 1 without any warning. The two thresholds are far from their populations, since genuine wells on the coarse penalty sit about 10^{-6} below the plateau and plateau ripple never reaches 10^{-2} of it. Rescue can only *add* a bracket, and the refiner discards a spurious one in a few evaluations. Any rescue marks the run floor-limited.

8.3 Outside this note's scope: stage policy

This note describes the search as v1 designed it: one objective, the pencil's threshold-rank tension, used by every stage. `chladni` also ships an opt-in variant, the flags `coarse_bracket` and `stable_rank` of `search.solve`, in which the stages read the pencil under different *policies* (`chladni.mps.policy`). Stage 1 brackets on a coarse penalty that makes no rank decision and is therefore continuous in λ. It runs with `rescue_floor=True` at a precision no tighter than $10\sqrt{\varepsilon_{\mathrm{m}}}$. Each bracket is then refined at the threshold rank first, and at a stable rank frozen over the root's cell when the root has a noise tail. Every algorithm in this note runs unchanged inside that variant. What the variant changes is the objective each stage is handed, and it is documented in `search.solve` and `notes/stage_policy_steps.md`, not here.

9 Refinement

The bracket is now handed to a refiner, which drives it to the requested accuracy.

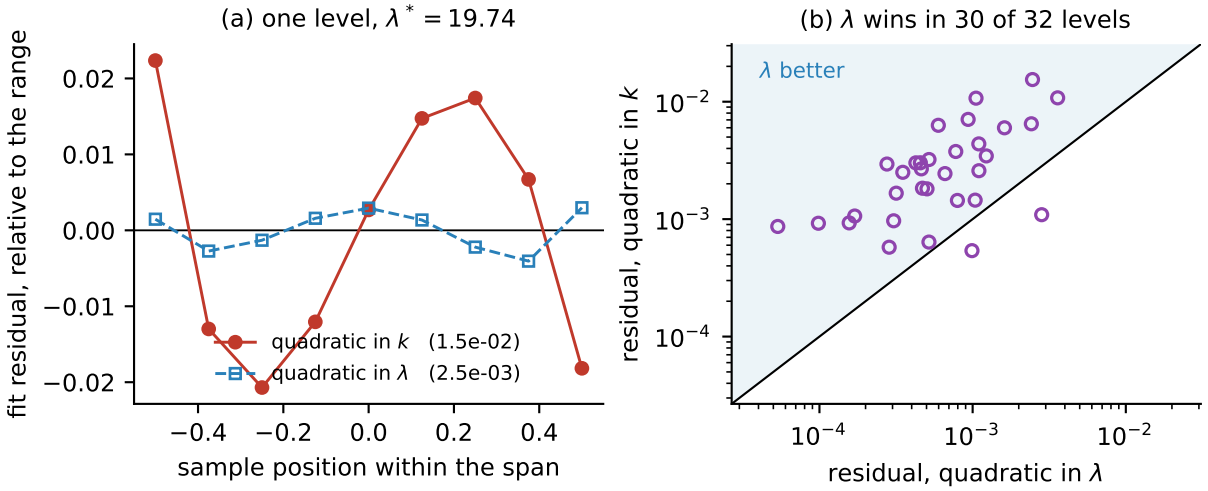


Figure 7: *(a)* Residuals of a quadratic fit to σ_1^2 at one level, over the same nine physical points, in k and in λ. The fit in k leaves a systematic W-shaped residual an order of magnitude larger. *(b)* The same comparison at every isolated level of four rectangles.

9.1 Iterated parabolic interpolation, not Brent

Because (17) is exactly quadratic, the right algorithm solves for the vertex from three points and iterates. Measured over 26 eigenvalues at prec $= 10^{-14}$, on brackets of width 10^{-6} to 10^{-1}:

refiner	median relative error	evaluations per bracket
iterated parabolic on σ_1^2	$\mathbf{1.6 \times 10^{-16}}$	**5.5–7.9**
Brent on σ_1^2	5.4×10^{-10} to 5.6×10^{-5}	5.0–11.7
straddled interpolation on σ_1	0 (exact)	48–63

Brent does not exploit the parabolic structure: its parabolic step is guarded by a golden-section fallback, and it terminates long before the curvature information is used up. A general-purpose scalar minimizer is the wrong tool here.

Remark 13 (Squaring does not cost half the significant digits). The familiar $\sqrt{\varepsilon_{\mathrm{m}}}$ limit — that a minimum can be located only to the square root of machine epsilon — belongs to methods that locate a minimum by *comparing function values*, where the quadratic flatness of the objective destroys the comparison. Solving for the vertex from the *curvature* does not suffer it: the information is in the second difference, not in which of two nearly equal values is smaller.

9.2 Refine in λ

§3 argued for k on the strength of the wall slope. The curvature says the opposite, and it governs here. Fitting a quadratic to σ_1^2 over *identical physical sample points*, the residual relative to the data range is smaller in λ than in k in 30 of 32 levels across four rectangles, with a median advantage of 4.4× in residual and 4.4× in the accuracy of the located vertex (Figure 7). The originating measurement reported 17 of 17 and a median of 5.1×.

This deserves an honest note, because it contradicts the model. Writing the cone as $\sigma_1 \approx \sqrt{\varepsilon^2 + c^2(k-k^*)^2}$ makes σ_1^2 exactly quadratic *in* k; the data say it is closer to quadratic in λ.

Fitting the model form directly leaves a residual of 1.5×10^{-2} against 8.7×10^{-4} for a plain quadratic in λ on the same points. So (2) is a good local model stated in a slightly wrong variable, and the two stages of the search are entitled to disagree about which variable they want because they are using different features of it.

9.3 Tolerance, and the guards

The stopping tolerance is

$$x_{\text{atol}} \;=\; \max\bigl(\text{prec} \cdot |\lambda|\,,\; 4\,\text{ulp}(\lambda)\bigr). \tag{18}$$

The second term is not decoration: a tolerance derived from λ alone can *exceed* the bracket it is refining — a 2.7×10^{-13} bracket handed a tolerance of 7.9×10^{-13} — so the refiner terminates before doing anything and returns a point that is not the minimum.

Four guards, each named with the failure it prevents.

1. **A parabola that never closed is a bad start before it is a verdict.** A non-positive leading coefficient means the three points do not describe a single minimum. The bracket may be empty, or it may hold two eigenvalues with the search's deepest sample on the hump between them, so that the fit is across a W. Refusing at once was measured to lose both members of such a pair: `solve` read the refusal as an empty bracket and still certified the count. So with no progress yet the refiner bisects the wider gap beside its lowest sample, toward a well if there is one, and tries the parabola again. It does this at most four times (`_FALLBACK_STEPS`) and then refuses with `no_parabola`. A lower sample found this way does not count as progress, so guard 2 can still tell the two cases apart. An empty bracket still walks to an endpoint within four steps and is refused there.

2. **After progress, the same condition means success.** It also fires at the *end* of a good refinement, once the three points have collapsed onto the floor and the curvature is gone. Treating that as failure discards correct answers.

3. **Track the lowest point seen independently.** The triple is re-formed by position rather than by value, so a bad vertex can otherwise discard the sub-interval holding the minimum.

4. **Accept only below both sampled ends.** The accepted point must be strictly interior (a margin of four ulps) and must have σ_1 strictly below its value at *both* ends of the bracket it was handed. Those ends were evaluated on the way in, so the test costs nothing. This is "compare against the cone outside the bracket, where it must be rising", read off the points the bracket already carries. An earlier version probed the cone at $\lambda \pm 4\cdot$width instead. On an empty bracket one wall away from a root, that probe jumped over the eigenvalue onto the plateau beyond, read "rising" on both sides, and accepted a wall point. §9.4 generalizes the lesson, and §9.5 covers the one case the ends cannot judge.

Two details of the iteration also matter for its accuracy.

- **The vertex is centred on the middle point.** The monomial form $-c_1/(2c_2)$ squares each abscissa about the origin while the samples sit near λ. Each square then carries its own $\text{ulp}(\lambda^2)$, and the vertex stalls at a spacing of about $\lambda\sqrt{\varepsilon_{\text{m}}}$. Under that form the refiner reported convergence at 10^{-11} to 10^{-9} on pencils whose floor is 10^{-15}. Written in offsets from the middle point, the error is a few ulps of λ at every spacing. The locator of §7 uses the same routine.

- **Fence, then stop.** Iterated vertices approach from one side, exactly as Remark 9 says, so a triple left to narrow by itself costs a halving per step. Once the vertex agrees with the best point to within $0.05\,x_{\text{atol}}$ (`_SETTLE`), the refiner instead samples at $\pm 0.45\,x_{\text{atol}}$ on each side not already fenced. That costs at most two evaluations, and the unimodal minimum is then bounded by a triple narrower than the tolerance. The remaining disagreement is carried into the reported bound, so this setting changes cost and never honesty.

9.4 The interiority trap

The last guard has a subtlety that produced the same bug twice, in two forms, and it generalizes past this algorithm.

The naive version demands that an accepted minimum be strictly interior to its bracket. But a bracket is guaranteed to *contain* the eigenvalue, not to have the eigenvalue comfortably inside it: the cover loop stops on purity-by-width, and a well 10^{-7} wide can sit inside an 8.6×10^{-7} bracket with its minimum very near an end. An interior-minimum rule discarded 8 of 26 correct brackets.

The refined version of the same error is subtler. Probe interiority by evaluating at $x \pm h$ for a *fixed* relative step $h = 10^{-9}k$ and demanding both exceed $f(x)$. This demonstrates interiority only when the located root is already nearer the true minimum than h is; otherwise one step moves *toward* the minimum, reads lower, and the genuine root is refused. On an L-shape window this threw away all three eigenvalues at the default setting, every one of which had been located to at worst 1.2×10^{-8}.

The right way to state the finding is as a separation rather than a threshold. The kept and dropped populations split by a factor of eight in the ratio (location error)$/h$ — 0.000 to 0.098 against 0.782 to 2.974 — but *not* at 1, because the probe compares tension values on flanks of unequal slope. A first version of the corresponding test asserted error $> h$ exactly and was falsified by the row at 0.78; what survives is the mechanism and the separation, not a constant.

Remark 14. The lesson generalizes, and it is the one to take away from this section. A probe whose reach is set by the bracket width cannot know what it is stepping across. A probe whose reach is set by a *constant* cannot know that either, and additionally cannot know how accurate the point it is stepping from is. The acceptance test that survives compares the candidate against the two *sampled* endpoints it already has, and asks nothing new of the objective — everywhere except at an end, where §9.5 sets the reach from the point's own cone.

9.5 The end probe

Guard 4 judges a point by the bracket's ends, and that is sound only when the ends are real. They are not quite real. The search hands the refiner a bracket in k, and converting it to λ widens each end by four ulps (§3). Four ulps is also the interior margin, so a sample sitting *exactly on* the search's bracket end is three or four ulps inside the widened one, and it passes as interior by rounding alone. That configuration is not exotic. The deepest sample the search holds can *be* its bracket end, lying on the wall of a *neighbouring* well, and the refiner then collapses onto it. Measured at $130\pi^2$, that phantom sat 0.48 licences from the eigenvalue and was accepted as a root. Position cannot separate this case from a genuine eigenvalue lying at the search's end, which an evenly spaced cluster at prec $= 10^{-14}$ produces, and neither can provenance: a test keyed on the point being bit-identical to a handed-in end was defeated by a vertex landing a tenth of an ulp away. What differs is the curve *beyond* the end. So the refiner is told the search's own bracket

(`refine(..., interior=)`, which every call in `search` passes), and `refine._end_verdict` looks there, spending evaluations only when the following test fires.

When a point is "at an end". The question concerns the point's resolution radius, not ulps. On the cone the minimum lies about σ/c from a point of tension σ, so the best point counts as at an end when that radius reaches the search's end, or when it is within the four-ulp tie. A tie alone let a phantom four ulps inside the upper end through at $125\pi^2$, as a fifth level of a four-member cluster, and its radius reached 0.7 bracket widths past the end. The slope c is the *steepest* secant from the best point to any sample at least a quarter of the search bracket away, and both qualifiers are measured.

- The fitted curvature of the last triple is the wrong slope. That triple sits in the ulp-scale noise pressed against the end, and its curvature read a slope 25 times too large.
- The secant to the far end alone is also the wrong slope. It assumes the phantom's bracket lies on one monotone wall. A bracket spanning a hump read a slope 18 times too shallow and overshot the neighbour. Samples past a hump lie below the wall, so they can only make a secant shallower, and the steepest secant is therefore the wall's.

Outward. The refiner evaluates at σ/c, then $\frac{1}{2}\sigma/c$, then $\frac{1}{4}\sigma/c$ beyond the end. These evaluations go straight to the objective rather than through the refiner's clamped evaluator, because a first version clamped the probe back onto the widened end, read the same value, and passed the phantom. The search stops at the first probe that reads below half the point's tension (`_DESCENT_RATIO`, the same value as the confirmation step's dip ratio). A falling curve alone does not prove a phantom, because a genuine root located to x_{atol} near the end also has a radius that crosses it, and its probe lands *on* the root. What separates them is where the minimum is. A probe falls only within about half a step of the minimum, so the point is ruled a phantom only when the falling probe lies more than half a step past the end. The halving covers a slope estimate up to six times too shallow. A fixed reach was measured to fail in both directions: one bracket width landed on the neighbouring well at $130\pi^2$ and overshot it at $25\pi^2$.

Inward. If nothing outside falls, the minimum is not outside, but it may be *inside* and unsampled. This happens when a bracket is nested in another: the deepest cached sample is the outer bracket's end, and the triple collapses there. At $50\pi^2$ this refused a member 1.6×10^{-7} inside a side bracket, and at $170\pi^2$ it accepted a root 2×10^{-7} from its member. So the refiner probes inward at the same three steps. The first probe lower than the end starts *one* nested refinement over the rest of the bracket, and if that refinement accepts, its result replaces this one. The nesting goes no deeper than one level.

Remark 15 (What the probe assumes). The probe is aimed by the cone. A well with a flat bottom — produced so far only by a penalized objective whose floor approaches the licence, never by the threshold — defeats it, because the steep side of a hump sets c and the reach then falls short of the gentle descent toward the neighbour. This is recorded as open work and does not affect the shipped default.

10 Cost, and the shape of the claim

10.1 Cost

Including the refinement each bracket hands on, at prec $= 10^{-8}$ on real pencils:

window	eigenvalues	evaluations per eigenvalue
a resolvable pair, $\lambda \in [10, 88]$	4	27
a multiplicity-3 cluster at $50\pi^2 \pm 3\%$	3	27
a wide window, $\lambda \in [10, 520]$	35	26

Roughly 26 evaluations per eigenvalue, near-flat in window size and in prec from 10^{-3} to 10^{-14}. For comparison, a uniform grid dense enough to detect everything costs about 43 per eigenvalue *and fails purity*, and grid-then-bisect costs 74–95 *and loses a third of the eigenvalues.*

10.2 The claim, stated conditionally

By Proposition 2 the completeness claim is not a theorem, and it must be reported as what it is:

> No x in $[a, b]$ with $\sigma_1(x) \le \tau$ exists outside the returned set, *given* the Lipschitz constant L.

Every run therefore reports L, the largest observed secant, the ratio between them, the number of times L grew after being relied upon, and the evaluation count. A large margin means the scan was conservative; a small one means the constant was being pushed and the run deserves a rerun at a larger safety factor. This matters because the failure mode is silent: a dip that is never sampled is invisible to any finite-evaluation method, and the exclusion search in particular returns nothing after its seed, with no error at all.

10.3 Declining is a third outcome

A method may assume the objective is well behaved at the scales its precision makes relevant, *detect that it is not*, and stop. Two triggers: the seed's slope estimate never settles (§4.3), or a long run of samples buys no reduction in the unresolved regions.

Declining is neither a pass nor a failure, and it is only honest if reported. A method that declines whenever it is about to be wrong has a perfect record over what it accepted and has demonstrated nothing, so the decline rate is quoted beside the pass rate over the same case set, always.

There is a related and easily conflated case which is *not* a decline: being asked for more precision than the objective can deliver. If the pencil's floor is binding, the right answer is to emit the brackets found, flag them floor-limited, and report the attained precision beside the requested one — not to report that nothing was found.

10.4 The band

The method's real limitation is best stated as a band. Write d for the separation of two eigenvalues at level x. Then

$$\frac{d}{x} < \text{prec} \quad \text{(merging is licensed)}, \qquad \frac{d}{x} > \text{the search's resolution} \quad \text{(must split, and can)},$$

and in between lies prec $< d/x <$ resolution: *must split, and cannot.* The band's width is the ratio of the resolution to prec, so

> the harder the precision requested, the wider the band of separations that cannot be handled.

This is a property of the configuration rather than a tuning failure, and it is the reason the honesty machinery of this section is not decoration. A method whose limits widen as it is asked to do better has to be able to say so.

11 What changed in the port

The algorithm was designed and measured in a standalone repository against a single family — the $(1 + \delta) \times 1$ rectangle with a plane-wave basis — and then ported into `chladni`. This section records the differences, because the gap between what was intended and what the implementation found is usually the most useful part.

11.1 What did not change

Almost all of the tuned constants, which is the more surprising result: safety factor 1.5, straddle 0.02, 1.5 samples per expected gap, a 10% settling tolerance, a 64-point seed without a Weyl hint, six doublings, the stall and locator caps at 200, the dip ratio 0.5, three confirmation evaluations, five cover/locate rounds, an isolation clearance of four bracket widths, and a gate margin of 1.5. The port is a set of *structural* corrections, not a retuning.

Remark 16. The gate margin deserves a footnote, because it invites a misreading: the originating module's own constant reads 2.0, but the searcher that uses it already defaulted to 1.5. The constant that ships is unchanged.

11.2 Genuine differences

	original	`chladni` v1, and why
architecture	an append-only `Evidence` store; every component a pure function of it, returning `NeedSample(x)` rather than evaluating	flat sample arrays with cover/locate/confirm in rounds, and a view object sharing the caller's cache at the $\lambda \leftrightarrow k$ seam. Offline testability traded for fitting the project's stage seams and a single shared, counted objective.
vertex	the parabolic vertex through the raw triple	solves for the quadratic coefficients by divided differences and *requires the leading coefficient to be positive*: a non-convex fit means the triple does not describe a single minimum.

	original	`chladni` v1, and why
fallback when the fit fails	sample the prediction, or the triple's midpoint	**bisect the wider side**. Using the straddle as a fallback shrinks the bracket by 2% per evaluation — measured creeping $6.2 \times 10^{-8} \to 3.4 \times 10^{-8}$ over twelve evaluations, then stopping.
straddle side	gaps to the nearest evidence samples	gaps to the certificate triple's own endpoints.
multiplicity reading in the gate	a parameter, defaulted to off	removed outright: a knob whose only safe setting is "off" is not a knob; see (16).
slope-growth cap	4, then decline	16. Growth is *cost, not error*, because exclusions are recomputed from the current L every pass. On a multiplicity-3 cluster the seed settles at 0.31 against a true wall slope of 0.683 and needs 7 growths to reach the right answer; at 4 it declined having found nothing, at 8 it passes at 80 evaluations.
exhaustion	abort	emit what is held, flag it floor-limited, and report the attained precision beside the requested one.
budget	an in-flight cap on a counted objective	600 evaluations *per expected eigenvalue*, never flat. A flat cap silently penalises dense windows, and once turned a 38-evaluation miss into the recorded claim that a whole family was unhandleable.
diagnostics	version-tagged benchmark tables	required output: L, its history, growths after use, the largest secant, attained against requested precision. The completeness claim is conditional on L, so L must be reported.
refiner start	—	the deepest *cached* sample rather than the arithmetic midpoint. A bracket is guaranteed to contain the eigenvalue, not to have a sampled descent inside it; this moved a located value from 1.8×10^{-10} to 7.4×10^{-12}.
refiner interior margin	—	removed (§9.4). It refused genuine off-centre minima — a level 8.8×10^{-8} from a bracket end against a demanded 2.15×10^{-7} — and was redundant with the test that the best value beat both ends.

11.3 Defects the port introduced, and what caught them

These are not differences of design; they are bugs the reimplementation created and the test tiers found. They are listed because each was found *by running, not by reading*, and because each is a plausible thing to do again.

- The budget defaulted to one eigenvalue where the objective offered no Weyl count, so the locator exhausted it and returned partial results *silently*.
- Regions were retired by overlap, so bracketing one member of a pair retired the whole region and lost the other — coverage 0.75 on every pair, which is §6's own failure mode arriving through the bookkeeping.
- The fast and slow paths were collapsed. The straddle places its sample at a predicted vertex, so on a region holding two wells it piles samples onto one and never lands between them: coverage 1.00 with purity failing at every precision. A region the gate has not certified must go back to the exclusion ladder.
- "Silence is not permission" applies on the way *out* of the locator too: with the post-check unable to certify, keeping the narrowed bracket discarded whatever else the region held.
- A region lying entirely *inside* a well reads as flat — every sample on the floor, internal ratio ≈ 1 — and was dropped by the confirmation step. Flatness at the floor is the opposite of flatness on the plateau, which is why the level test is a fraction of the measured plateau (Remark 2).
- The loop budget counted cache *misses*, so a refiner re-reading known points spun forever at zero cost.

11.4 Three stopping rules kept as comments

The implementation records three rules that were tried and removed, each with the number that removed it: a positive threshold on the envelope (broke 66 of 66 cases, Remark 5); a flat-bottom stop derived from $4\sigma_{\min}/c$, whose estimate is an *upper* bound on the floor and so stopped narrowing far too early; and a prominence stop on the certificate triple, which is sound in principle and unsafe early for the same reason the further-flank normalization exists. Recording a rejected idea beside the code, with its measurement, is what makes the design legible; it is also why the list in §11 could be written at all.

12 What is derived and what is measured

A reader who takes one thing from this note should take the difference between these two columns, because the algorithm's constants were all measured on *one* family and several still carry recalibration notes.

Follows from the model	**Measured, and may not transfer**
$p = 1/2$ is the unique power making the wall slope level-independent, (5)	that the spread in k is as small as 5–6%, and the value $C \approx 0.27$
the envelope and its minimizer, Lemma 1	that L from the largest secant is nearly tight
the ladder ratio $(s+1)/(s-1)$, independent of c, Proposition 3	safety $s = 1.5$, gate margin 1.5, straddle 0.02
σ_1^2 quadratic $\Rightarrow$ an exact vertex from three points	that λ beats k for the fit (30 of 32 here, 17 of 17 there)
the precision factor $\mathrm{prec}_u = p \cdot \mathrm{prec}$, (6)	$\sim$26 evaluations per eigenvalue
the cancellation-free c, (15)	that σ_2/c is accurate to 3% where a neighbour is close
$\sigma_2/\sigma_1 = 1 + d/u$, (16)	the $\lvert\Omega\rvert \,/\, \lvert\partial\Omega\rvert$ constant $\kappa \approx 2.6$

The originating design's own summary of its position is worth quoting, because it is the right posture toward every number above: the largest uncertainty is not in any of the individual measurements, but in the fact that all of them come from one rectangle. Several have since been re-measured on the disk, the equilateral triangle and the L-shape; the ones that have not are the ones to check first.

Sources

The design records are `docs/design/CHLADNI_V1.md` (the bracketing spec and its as-built deltas), `docs/design/search.md` (the precision contract, the ground-truth definition, and the multiplicity and collapse rules not developed here) and `docs/design/candidates.md` (the stage seams and the comparison protocol). The implementation is `src/chladni/mps/bracket.py` (stage 1, `bracket_interval`) and `src/chladni/mps/refine.py` (stage 2, `refine`), composed with the multiplicity stage by `src/chladni/mps/search.py` (`solve`). Both stages were promoted from candidates in `tests/` after they won on the scored seam. `tests/candidates_v1.py` keeps the adapters that still score them there against the baselines. Figures are regenerated by `docs/figures/make_bracket_figures.py`, which drives the same code the text describes and prints every number quoted in a caption.

Not developed here, and specified in `search.md`: the multiplicity stage, which reads σ_{j+1}/σ_j against a wide bracket at zero additional cost; the collapse-versus-merge rule for roots reported close together; and the prec-indexed ground truth $S(\mathrm{prec})$ against which all of it is scored.